In[ ]:=

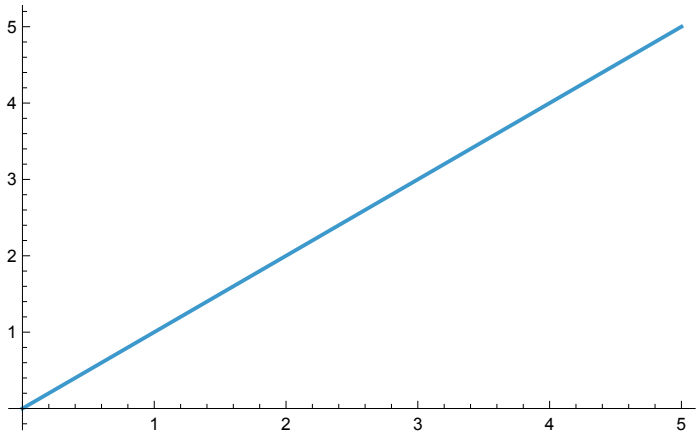
Out[ ]=

In[ ]:=

Problem 1) Plot y = x for 0 < x < 5. Start with the Plot command. If you don't know where to begin,
type "Plot" into Mathematica, then right click what you've typed and select "Get Help."

In[ ]:=
```
Plot [x, {x, 0, 5}]
```

Out[ ]=



Problem 2) Use Mathematica to find the first and second derivatives of f (t) = α ln(at) cos(ωt) sin(ωt).

In[ ]:=
```
Clear[f, t, a, α, ω]
f [t_] := α Log[a t] Cos[ω t] Sin[ω t]
f'[t]
f''[t]
```

Out[ ]=

$$\alpha\,\omega\,\text{Cos}[t\,\omega]^2\,\text{Log}[a\,t] + \frac{\alpha\,\text{Cos}[t\,\omega]\,\text{Sin}[t\,\omega]}{t} - \alpha\,\omega\,\text{Log}[a\,t]\,\text{Sin}[t\,\omega]^2$$

Out[ ]=

$$\frac{2\,\alpha\,\omega\,\text{Cos}[t\,\omega]^2}{t} - \frac{\alpha\,\text{Cos}[t\,\omega]\,\text{Sin}[t\,\omega]}{t^2} - 4\,\alpha\,\omega^2\,\text{Cos}[t\,\omega]\,\text{Log}[a\,t]\,\text{Sin}[t\,\omega] - \frac{2\,\alpha\,\omega\,\text{Sin}[t\,\omega]^2}{t}$$

Problem 3)Use Mathematica to find the derivative with respect to x of the arbitrary function w(x) =
f (x) g(x) h(x). Hint: you can input an arbitrary function as f [x] in Mathematica . To start,
try taking the derivative of just f [x]

In[•]:=
```
Clear[f, g, h, w, x]
```

In[•]:=
```
w[x_] := f[x] × g[x] × h[x]
w'[x]
```

Out[•]=

$$g[x] \times h[x]\ f'[x] + f[x] \times h[x]\ g'[x] + f[x] \times g[x]\ h'[x]$$

Problem 4)

Use the N DSolve[] function to solve our Equation of Motion for the pendulum oscillator:

..ϕ = –ω2

0 sin(ϕ) (1)

where ω0 = pg/*l*. Use *l* = 10 cm, ˙ ϕ(t = 0) = 0 and ϕ(t = 0) = 30○. Plot your solution, and be sure to refer to the documentation for this: it has a number of great examples. This

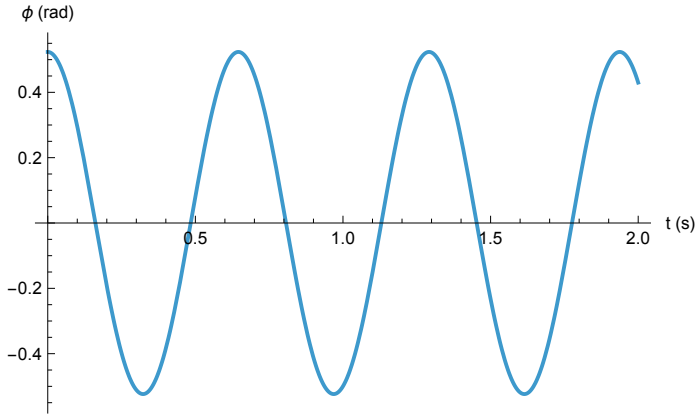
should be much easier than the python version of this exercise

In[ ]:=
```
Clear[g, l, ω0, ϕ, t, sol]
g = 980;  (*cm/s^2*)
l = 10;  (*cm*)
ω0 = Sqrt[g / l];

sol = NDSolve[{ϕ''[t] == -ω0^2 Sin[ϕ[t]], ϕ[0] == 30 Degree, ϕ'[0] == 0}, ϕ, {t, 0, 2}];

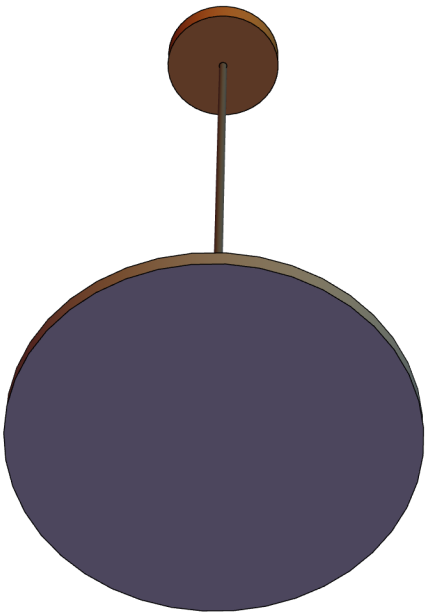
Plot[Evaluate[ϕ[t] /. sol], {t, 0, 2}, AxesLabel → {"t (s)", "ϕ (rad)"}]
```

Out[ ]=



In[7]:=
```
Graphics3D[{Gray, Cylinder[{{0, 0, 0}, {0, 0, 0.3}}, 3],
  Darker[Gray], Cylinder[{{0, 0, 0.3}, {0, 0, 12}}, 0.07],
  Brown, Cylinder[{{0, 0, 12}, {0, 0, 12.4}}, 1]},
 Boxed → False, Axes → False]
```

Out[•]=



In[•]:=
```
Show[Graphics3D[{Gray, Cylinder[{{0, 0, 0}, {0, 0, 0.3}}, 3],
    Darker[Gray], Cylinder[{{0, 0, 0.3}, {0, 0, 12}}, 0.07],
    Brown, Cylinder[{{0, 0, 12}, {0, 0, 12.4}}, 1]},
  Boxed → False, Axes → False], Axes → True, AxesStyle → Gray, Boxed → True]
```

Out[10]=

